ORDER-ZERO OPERATORS AND IRREDUCIBILITY OF POWER-RELATED MATRIX WEIGHTS

KNOWLEDGE SENSE

ABSTRACT. Weight matrices $W = T\,\mathrm{diag}(w_1, \dots, w_N)T^*$ of every size N, with $T(x) = I + xA$ for an explicit nilpotent matrix A depending on nonzero real parameters, were introduced by Bono Parisi and Pacharoni for certain classical scalar weights and studied by Bono Parisi for arbitrary scalar weights. Bono Parisi conjectured that, when the scalar weights are related by powers of x, the algebra $\mathcal{D}(W)$ contains only scalar operators of order zero, and he stated that this holds for $N \le 10$. The power relation can be read in two ways, giving $w_j = w\,x^{s_j}$ with $(s_j) = (0, 0, 1, \dots, N-2)$ or $(s_j) = (0, 1, \dots, N-1)$. We prove the conjecture for both exponent lists and every N. More precisely, for all nonzero real parameters and every admissible scalar weight w on an interval, bounded or unbounded, the operators of order zero in $\mathcal{D}(W)$ are the complex multiples of the identity, and W is irreducible. Since the interval may be unbounded, we show directly that for weights $W = wP$, with w a scalar weight and P a polynomial matrix, the existence of a non-scalar operator of order zero implies the existence of a non-scalar W-symmetric constant matrix. As a consequence, the Laguerre-type examples give, in every size, irreducible weight matrices that are Darboux transformations of diagonal scalar weights and whose orthogonal polynomials satisfy a second-order differential equation.

1. INTRODUCTION

A weight matrix of size N on an interval $\Delta = (x_0, x_1)$, with $-\infty \le x_0 < x_1 \le \infty$, is an integrable function $W\colon \mathbb{R} \to \mathrm{Mat}_N(\mathbb{C})$ that is positive definite almost everywhere on Δ, vanishes outside Δ, and has finite moments of all orders. For a sequence $(Q_n)_{n\ge 0}$ of matrix-valued orthogonal polynomials for W, the algebra $\mathcal{D}(W)$ consists of the differential operators $D = \sum_{j=0}^{m} \partial^j F_j(x)$, acting on the right by $P \cdot D = \sum_j \partial^j(P)F_j$, such that $Q_n \cdot D = \Lambda_n(D)Q_n$ with $\Lambda_n(D) \in \mathrm{Mat}_N(\mathbb{C})$ for every n [3], [1, Definition 2.2]; it does not depend on the choice of the sequence. A weight matrix is *reducible* if there are a nonsingular $M \in \mathrm{Mat}_N(\mathbb{C})$ and weight matrices W_1, W_2 of positive sizes with $W = M\,\mathrm{diag}(W_1, W_2)M^*$ on Δ, and *irreducible* otherwise [1, Definition 2.4]. Tirao and Zurrián [6] showed that W is reducible if

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and only if the real vector space

$$\mathcal{C}_{\mathbb{R}}(W) = \{F \in \mathrm{Mat}_N(\mathbb{C})\colon FW(x) = W(x)F^* \text{ for almost every } x \in \Delta\}$$

of W-*symmetric* matrices is larger than $\mathbb{R}I$, and that a reducible weight has non-scalar operators of order zero in $\mathcal{D}(W)$; the converse part of [6, Theorem 4.3] assumes a bounded interval.

Let $N \geq 1$ and let $a_1, \dots, a_{N-1}$ be nonzero real numbers. With the standard matrix units E_{ij} put

$$A = \sum_{r=1}^{\lfloor N/2 \rfloor} a_{2r-1} E_{2r-1,2r} + \sum_{r=1}^{\lfloor (N-1)/2 \rfloor} a_{2r} E_{2r+1,2r}, \tag{1.1}$$

and $T(x) = e^{xA}$ for real x. Every nonzero entry of A has an odd row index and an even column index, so $A^2 = 0$, $T(x) = I + xA$ and $T(x)^{-1} = I - xA$. Weight matrices of the form

$$W(x) = T(x)\,\mathrm{diag}\big(w_1(x), \dots, w_N(x)\big)T(x)^* \tag{1.2}$$

were introduced by Bono Parisi and Pacharoni [2, Definition 1.1] for classical scalar weights of the same type whose quotients w_i/w_j are not rational functions. Bono Parisi [1] studied (1.2) for arbitrary scalar weights $w_1, \dots, w_N$ on a common interval. He gave explicit orthogonal polynomials for W, sufficient conditions for bispectrality, and several irreducibility results. For $N = 2$ the weight reduces if and only if $w_1(x) = -a_1^2 w_2(x)(b-x)(c-x)$ for some real $b < c$ with $\Delta \subseteq (b, c)$ [1, Proposition 5.1]; for arbitrary N it is irreducible when no quotient w_i/w_j with $i \neq j$ is a rational function [1, Proposition 5.6], a statement attributed there to [2, Proposition 3.7], where it is proved for the Hermite-, Laguerre- and Jacobi-type families of [2, Definition 1.1]. At the end of [1, Section 5] the weights are related instead by powers of x, $w_{i+1}(x) = w_1(x)x^{i-1}$, so that all these quotients are rational. It is stated there that for $N \leq 10$ the algebra $\mathcal{D}(W)$ contains only scalar operators of order zero, and conjectured that this holds for arbitrary N. Read with $i = 1, \dots, N-1$, the displayed relation gives the exponents $(0, 0, 1, \dots, N-2)$, whereas the 10×10 Laguerre-type example that follows it, with $w_j(x) = e^{-x}x^{\alpha+j-1}$, has the exponents $(0, 1, \dots, 9)$. We prove the conjecture for both exponent lists.

For $N \geq 1$ let

$$\mathsf{L}_N = (0, 0, 1, 2, \dots, N-2) \ \ (N \geq 2), \quad \mathsf{L}_1 = (0), \quad \mathsf{E}_N = (0, 1, 2, \dots, N-1).$$

A scalar weight is a weight matrix of size one. For a list $s = (s_1, \dots, s_N)$ of nonnegative integers we call a scalar weight w *s-admissible* on Δ if $w(x)x^{s_j}$ is a scalar weight on Δ for every j (with $x^0 = 1$). We put

$$\begin{gathered} D_s(x) = \mathrm{diag}(x^{s_1}, \dots, x^{s_N}), \qquad P_s(x) = T(x)D_s(x)T(x)^*, \\ W_s(x) = w(x)P_s(x). \end{gathered} \tag{1.3}$$

Thus W_s is the weight (1.2) with $w_j = w\,x^{s_j}$. Since $s_1 = 0$ for both exponent lists, w itself is a scalar weight. If s contains an odd exponent, which happens

for E_N with $N \geq 2$ and for L_N with $N \geq 3$, admissibility forces $x_0 \geq 0$; for $\mathsf{L}_2 = (0, 0)$ and $N = 1$ any interval is possible. In all cases W_s is a weight matrix on Δ: it is positive definite wherever all $w(x)x^{s_j}$ are positive, because $T(x)$ is invertible, and its entries are w times polynomials. Write $\mathcal{D}_0(W)$ for the set of operators of order zero in $\mathcal{D}(W)$.

Theorem 1.1. *Let $N \geq 1$, let $a_1, \ldots, a_{N-1}$ be nonzero real numbers, let $s \in \{\mathsf{L}_N, \mathsf{E}_N\}$, and let w be an s-admissible scalar weight on an interval Δ, bounded or not. Then:*

(a) $\mathcal{C}_{\mathbb{R}}(W_s) = \mathbb{R}I$;
(b) *W_s is irreducible;*
(c) $\mathcal{D}_0(W_s) = \mathbb{C}I$, *that is, every operator of order zero in $\mathcal{D}(W_s)$ is a complex multiple of the identity.*

Thus the conjecture at the end of [1, Section 5] holds for every N and for both exponent lists.

Corollary 1.2. *Let $N \geq 1$, let $a_1, \ldots, a_{N-1}$ be nonzero real numbers, and let $w_1, \ldots, w_N$ be scalar weights on a common interval with either $w_{i+1}(x) = w_1(x)x^{i-1}$ for $1 \leq i \leq N-1$, or $w_{i+1}(x) = w_1(x)x^i$ for $1 \leq i \leq N-1$. Then the weight matrix* (1.2) *is irreducible and $\mathcal{D}(W)$ contains only scalar operators of order zero.*

Combined with the Laguerre-type results of [1, Section 4.1], Theorem 1.1 gives irreducible examples of every size with the following properties.

Corollary 1.3. *Let $N \geq 1$, $\alpha > -1$, $s \in \{\mathsf{L}_N, \mathsf{E}_N\}$, and let $a_1, \ldots, a_{N-1}$ be nonzero real numbers. The weight matrix*

$$W(x) = T(x)\,\mathrm{diag}\big(e^{-x}x^{\alpha+s_1}, \ldots, e^{-x}x^{\alpha+s_N}\big)T(x)^*, \qquad x \in (0, \infty),$$

is irreducible and satisfies $\mathcal{D}_0(W) = \mathbb{C}I$. The orthogonal polynomials of W given in [1, Theorem 3.1] *are eigenfunctions of a second-order differential operator, and W is a Darboux transformation of* $\mathrm{diag}(e^{-x}x^{\alpha+s_1}, \ldots, e^{-x}x^{\alpha+s_N})$ *in the sense of* [1, Definition 2.6].

Proof. The function $w(x) = e^{-x}x^{\alpha}$ on $(0, \infty)$ is s-admissible because $\alpha + s_j > -1$, so the first assertion is Theorem 1.1. The second-order operator is the one constructed in [1, Section 4.1] from [1, Theorem 4.1], which applies to arbitrary Laguerre parameters. The parameters $\alpha + s_j$ have integral differences, so [1, Theorem 4.3] gives the Darboux transformation. □

For $N = 10$ and $s = \mathsf{E}_{10}$ this is the example at the end of [1, Section 5]. Irreducible weights (1.2) of every size were known from the criterion on non-rational quotients [1, Proposition 5.6]; for the Hermite-, Laguerre- and Jacobi-type families of [2, Definition 1.1] those weights are not Darboux transformations of direct sums of classical scalar weights in the sense of [2, Definition 2.5], by [2, Theorem 1.4]; see also [1, Section 4]. Irreducible Laguerre-type weights that are Darboux transformations of diagonal scalar weights in the sense of [1, Definition 2.6] were exhibited in [1] for $N = 5$,

with exponents $(0,0,1,1,2)$ [1, (4.3) and Section 5], and in the power-related cases stated for $N \le 10$; Corollary 1.3 provides them in every size.

Known and new parts. The construction (1.1)–(1.2) was introduced by Bono Parisi and Pacharoni [2] for the classical families of [2, Definition 1.1] and extended to arbitrary scalar weights by Bono Parisi [1]. The power-related weights, the conjecture and the statement for $N \le 10$ are due to Bono Parisi [1]; the case $N = 2$ of the irreducibility in (b) also follows from [1, Proposition 5.1]. The reducibility criterion that turns (a) into (b) is [6, Theorem 2.8, Corollary 2.11]. New in this paper are a uniform proof of (a)–(c) for all N and both exponent lists, which includes the sizes $N \le 10$ stated in [1], and the implication (a)⇒(c) on unbounded intervals for weights of the form $W = wP$ (Proposition 2.3). The latter is needed because the converse part of [6, Theorem 4.3] assumes a bounded interval, while the Laguerre-type weights live on $(0,\infty)$; we therefore prove the required implication directly for the weights $W = wP$ considered here. Our argument uses only that W_s is a scalar weight times a polynomial matrix, and assumes neither bounded support nor determinacy of the moment problem.

Method. The core of the proof is a direct computation of $\mathcal{C}_{\mathbb{R}}(W_s)$. Conjugating a W_s-symmetric matrix F by $T(x)$ and ordering the coordinates as odd indices followed by even indices, the condition $FW_s = W_sF^*$ becomes the identity $G(x)D_s(x) = D_s(x)G^*(x)$ between matrix polynomials, where G^* is the coefficientwise adjoint and the lower-left block of G is a constant matrix U. Degree comparisons, together with the full column rank of the rectangular block of A, force $U = 0$; the identity then forces G to be real diagonal, and the path structure of A forces all diagonal entries to coincide. The repeated exponent 0 in L_N needs one additional equation from the odd block.

Organization. Section 2 collects facts on operators of order zero and proves the implication (a)⇒(c) for weights of the form $W = wP$. Section 3 introduces the block form, and Section 4 proves Theorem 1.1.

2. Operators of order zero and W-symmetric matrices

Throughout this section W is a weight matrix of size N on an interval Δ, which may be unbounded, and $\langle P, Q\rangle = \int_\Delta P(x)W(x)Q(x)^*\,dx$ for matrix polynomials P, Q. If v is a nonzero row vector polynomial, then $v(x)W(x)v(x)^* > 0$ for almost every $x \in \Delta$, since v has finitely many zeros; hence $\int vWv^* > 0$. Thus the block Hankel matrices of the moments of W are positive definite, and there is a unique sequence $(Q_n)_{n\ge0}$ of monic orthogonal polynomials, $Q_n(x) = x^nI + (\text{lower terms})$ and $\langle Q_n, Q_m\rangle = 0$ for $n \ne m$; see [5] and [1, Section 2.1]. In particular $Q_0 = I$.

Lemma 2.1. *An operator $D = F_0(x)$ of order zero belongs to $\mathcal{D}(W)$ if and only if F_0 is a constant matrix C such that $CQ_n = Q_nC$ for every $n \ge 0$.*

Proof. Since $\mathcal{D}(W)$ does not depend on the sequence of orthogonal polynomials [1, Remark 2.3], we use (Q_n). If $D \in \mathcal{D}(W)$, then $Q_0 \cdot D = F_0(x) = \Lambda_0(D)$ is constant, say $F_0 = C$. For $n \geq 0$ the identity $Q_n(x)C = \Lambda_n(D)Q_n(x)$ gives $C = \Lambda_n(D)$ on comparing the coefficients of x^n. Conversely, if C commutes with every Q_n, then $Q_n \cdot C = CQ_n$, so $C \in \mathcal{D}(W)$ with $\Lambda_n(C) = C$. □

We identify $\mathcal{D}_0(W)$ with the algebra of constant matrices described in Lemma 2.1. The next lemma records the facts from [6] and [4] that we use; their proofs do not use the boundedness of Δ. Part (i) concerns only the values $W(x)$, and the proof of part (ii) uses only the integral inner product and orthogonality.

Lemma 2.2. *Let W be a weight matrix on an interval Δ, bounded or not.*

(i) *W is irreducible if and only if $\mathcal{C}_{\mathbb{R}}(W) = \mathbb{R}I$* [6, Theorem 2.8, Corollary 2.11].

(ii) $\mathcal{C}_{\mathbb{R}}(W) \subseteq \mathcal{D}_0(W)$ [6, Theorem 4.2], [4, Lemma 3.1].

Proof. Part (ii) is the inclusion proved in the first half of the proof of [6, Theorem 4.2]; we recall the argument. Let $F \in \mathcal{C}_{\mathbb{R}}(W)$, so that $\langle PF, Q\rangle = \langle P, QF\rangle$ for all matrix polynomials P, Q. Suppose that F commutes with Q_j for $j < n$. Then $\langle Q_nF - FQ_n, Q_j\rangle = \langle Q_n, Q_jF\rangle - F\langle Q_n, Q_j\rangle = \langle Q_n, FQ_j\rangle = 0$ for $j < n$. Since Q_n is monic, $Q_nF - FQ_n$ has degree less than n and is orthogonal to all polynomials of degree less than n; hence it vanishes. By induction F commutes with every Q_n, and $F \in \mathcal{D}_0(W)$ by Lemma 2.1. □

On a bounded interval the reverse implication, from a non-scalar element of $\mathcal{D}_0(W)$ to a non-scalar element of $\mathcal{C}_{\mathbb{R}}(W)$, is [6, Theorems 4.2 and 4.3]; its proof uses that a function on a bounded interval whose moments all vanish is zero. For the weights of this paper the following replaces that step.

Proposition 2.3. *Let $W = wP$, where w is a scalar weight on an interval Δ, possibly unbounded, and P is an $N \times N$ matrix polynomial with $P(x)^* = P(x)$ for real x, such that W is a weight matrix on Δ. Let $M_0 = \int_\Delta W(x)\,dx$ and $K = M_0^{-1/2}$. If $C \in \mathcal{D}_0(W)$, then KCK^{-1} commutes with $KP(x)K$ for every $x \in \mathbb{R}$. Consequently, $\mathcal{D}_0(W) = \mathbb{C}I$ if and only if $\mathcal{C}_{\mathbb{R}}(W) = \mathbb{R}I$.*

Proof. The matrix M_0 is positive definite by the remarks at the beginning of this section. Put $\widetilde{P} = KPK$ and $\widetilde{W} = KWK = w\widetilde{P}$, so that $\int_\Delta \widetilde{W} = I$. The polynomials $R_n = KQ_nK^{-1}$ are monic, and $\int_\Delta R_n\widetilde{W}R_m^* = K\langle Q_n, Q_m\rangle K$ vanishes for $n \neq m$; thus (R_n) is the monic orthogonal sequence of $\widetilde{W}$. The matrix $C' = KCK^{-1}$ commutes with every R_n, hence with every coefficient of R_n. Write $R_n(x) = \sum_{j=0}^n L_{n,j}x^j$ with $L_{n,n} = I$, and let $\widetilde{M}_k = \int_\Delta x^k\widetilde{W}(x)\,dx$. Orthogonality of R_n to $R_0 = I$ gives

$$\widetilde{M}_n = -\sum_{j=0}^{n-1} L_{n,j}\widetilde{M}_j \qquad (n \geq 1).$$

Since $\widetilde{M}_0 = I$, induction on n shows that C' commutes with every $\widetilde{M}_k$. Let z be an entry of the matrix polynomial $C'\widetilde{P} - \widetilde{P}C'$, say $z(x) = \sum_{j=0}^{d} c_j x^j$. Then

$$\int_\Delta x^k w(x) z(x)\, dx = 0 \qquad (k \geq 0), \tag{2.1}$$

because the left-hand side is an entry of $C'\widetilde{M}_k - \widetilde{M}_k C'$; all these integrals converge absolutely because w has finite moments. Multiplying (2.1) for $k = 0, \dots, d$ by $\overline{c_k}$ and adding, we obtain $\int_\Delta w(x)|z(x)|^2\, dx = 0$, where we used that x is real. As $w > 0$ almost everywhere on Δ, the polynomial z vanishes almost everywhere on the interval Δ and hence identically. This proves the first assertion.

If $\mathcal{C}_{\mathbb{R}}(W) = \mathbb{R}I$ and $C \in \mathcal{D}_0(W)$, then C' commutes with the Hermitian matrices $\widetilde{P}(x)$, $x \in \mathbb{R}$, and so does C'^*. Both $H_1 = (C' + C'^*)/2$ and $H_2 = (C' - C'^*)/(2i)$ are Hermitian and commute with every $\widetilde{W}(x)$. For $H \in \{H_1, H_2\}$ the matrix $F = K^{-1}HK$ satisfies $F^* = KHK^{-1}$ and therefore

$$FW = K^{-1}H\widetilde{W}K^{-1} = K^{-1}\widetilde{W}HK^{-1} = WF^*,$$

that is, $F \in \mathcal{C}_{\mathbb{R}}(W) = \mathbb{R}I$. Hence H_1 and H_2 are real multiples of I, and $C = K^{-1}(H_1 + iH_2)K \in \mathbb{C}I$. Conversely, if $\mathcal{D}_0(W) = \mathbb{C}I$, then Lemma 2.2(ii) gives $\mathcal{C}_{\mathbb{R}}(W) \subseteq \mathbb{C}I$, and $cW = W\overline{c}$ with $W \neq 0$ forces $c \in \mathbb{R}$. □

3. The block form

Let $N \geq 2$, put $p = \lceil N/2 \rceil$ and $q = \lfloor N/2 \rfloor$, and order the coordinates of $\mathbb{C}^N$ as the odd indices $1, 3, \dots, 2p-1$ followed by the even indices $2, 4, \dots, 2q$. This reordering is a conjugation by a permutation matrix; it preserves irreducibility and transforms $\mathcal{C}_{\mathbb{R}}(W)$ by the same conjugation, so we may work in the new coordinates. There

$$A = \begin{pmatrix} 0 & B \\ 0 & 0 \end{pmatrix}, \qquad T(x) = \begin{pmatrix} I_p & xB \\ 0 & I_q \end{pmatrix}, \qquad T(x)^{-1} = \begin{pmatrix} I_p & -xB \\ 0 & I_q \end{pmatrix},$$

where B is the real $p \times q$ matrix whose only nonzero entries are

$$B_{r,r} = a_{2r-1} \quad (1 \leq r \leq q), \qquad B_{r+1,r} = a_{2r} \quad (1 \leq r \leq q,\ r+1 \leq p). \tag{3.1}$$

Indeed, the entry a_{2r-1} of A sits in row $2r-1$ and column $2r$, and the entry a_{2r} sits in row $2r+1$ and column $2r$; the condition $r \leq \lfloor (N-1)/2 \rfloor$ in (1.1) is equivalent to $r + 1 \leq p$. In particular B is the biadjacency matrix of a path through all N coordinates, with nonzero weights.

Lemma 3.1. *The matrix B has rank q. Consequently $BY = 0$ implies $Y = 0$ for every complex matrix Y with q rows.*

Proof. By (3.1) the first q rows of B form a lower bidiagonal matrix with diagonal $a_1, a_3, \dots, a_{2q-1}$, whose determinant is nonzero. □

In the new coordinates $D_s = \operatorname{diag}(D_o, D_e)$ with $D_o = \operatorname{diag}(x^{o_1}, \dots, x^{o_p})$ and $D_e = \operatorname{diag}(x^{e_1}, \dots, x^{e_q})$, where $o_r = s_{2r-1}$ and $e_t = s_{2t}$. Explicitly,

$$\begin{aligned} s = \mathsf{E}_N\colon &\quad o_r = 2r-2, \qquad e_t = 2t-1; \\ s = \mathsf{L}_N\colon &\quad o_1 = 0, \quad o_r = 2r-3 \ (r \geq 2), \qquad e_t = 2t-2. \end{aligned} \tag{3.2}$$

Lemma 3.2. *For $s \in \{\mathsf{L}_N, \mathsf{E}_N\}$:*

(i) *$o_1, \dots, o_p$ are pairwise distinct, and so are $e_1, \dots, e_q$;*
(ii) *for $1 \leq r \leq p$ and $1 \leq t \leq q$ we have $o_r - e_t \in \{0,1,2\}$ if and only if either $r = t+1$, in which case $o_r - e_t = 1$, or $s = \mathsf{L}_N$ and $r = t = 1$, in which case $o_r - e_t = 0$.*

Proof. Part (i) is clear from (3.2). In both cases $o_r - e_t = 2(r-t) - 1$ unless $s = \mathsf{L}_N$ and $r = 1$; this odd number lies in $\{0,1,2\}$ exactly when $r = t+1$. If $s = \mathsf{L}_N$ and $r = 1$, then $o_1 - e_t = 2 - 2t$, which lies in $\{0,1,2\}$ exactly when $t = 1$. □

Let $F \in \operatorname{Mat}_N(\mathbb{C})$, written in blocks of sizes p and q as

$$F = \begin{pmatrix} R & Y \\ U & S \end{pmatrix}.$$

Multiplying the triangular block matrices gives

$$G(x) = T(x)^{-1} F T(x) = \begin{pmatrix} R - xBU & Y + x(RB - BS) - x^2 BUB \\ U & S + xUB \end{pmatrix}. \tag{3.3}$$

For a matrix polynomial $Z(x) = \sum_j Z_j x^j$ we write $Z^*(x) = \sum_j Z_j^* x^j$, so that $Z^*(x) = Z(x)^*$ for real x.

Lemma 3.3. *If $F \in \mathcal{C}_{\mathbb{R}}(W_s)$, then*

$$G(x) D_s(x) = D_s(x) G^*(x) \tag{3.4}$$

as an identity of matrix polynomials.

Proof. Since $w > 0$ almost everywhere on Δ, the relation $FW_s = W_s F^*$ gives $FP_s(x) = P_s(x) F^*$ for almost every $x \in \Delta$. Both sides are matrix polynomials, and a polynomial vanishing on a set of positive measure is zero, so this is an identity of matrix polynomials. Since T has real coefficients, $T^* = T^{\mathsf{T}}$ and $(T^{-1} F T)^* = T^* F^* (T^*)^{-1}$; multiplying the identity on the left by T^{-1} and on the right by $(T^*)^{-1}$ gives (3.4). □

In blocks, (3.4) consists of the three identities

$$(R - xBU) D_o = D_o (R - xBU)^*, \tag{3.5}$$

$$\big(Y + x(RB - BS) - x^2 BUB\big) D_e = D_o U^*, \tag{3.6}$$

$$(S + xUB) D_e = D_e (S + xUB)^*, \tag{3.7}$$

the fourth block being the adjoint of (3.6).

4. Proof of Theorem 1.1

Proposition 4.1. *Under the hypotheses of Theorem 1.1, $\mathcal{C}_{\mathbb{R}}(W_s) = \mathbb{R}I$.*

We first treat $N = 1$. Then $W_s = w$ is scalar, and $cw = w\overline{c}$ forces $c \in \mathbb{R}$, so $\mathcal{C}_{\mathbb{R}}(W_s) = \mathbb{R}I$. From now on let $N \geq 2$ and $F \in \mathcal{C}_{\mathbb{R}}(W_s)$, with the notation of Section 3. We show that the lower-left block of G vanishes.

Lemma 4.2. $U = 0$.

Proof. Write $G_{12} = Y + x(RB - BS) - x^2BUB$ for the upper-right block of G, a matrix polynomial of degree at most two. The (r,t) entry of (3.6) reads

$$(G_{12})_{r,t}(x)\, x^{e_t} = \overline{U_{t,r}}\, x^{o_r}. \tag{4.1}$$

If $U_{t,r} \neq 0$, the right-hand side is a nonzero monomial divisible by x^{e_t}, so $o_r \geq e_t$ and $(G_{12})_{r,t} = \overline{U_{t,r}} x^{o_r - e_t}$; as G_{12} has degree at most two, $o_r - e_t \in \{0,1,2\}$. By Lemma 3.2(ii) the only entries of U that can be nonzero are

$$u_t = U_{t,t+1} \quad (1 \leq t \leq q,\ t+1 \leq p), \qquad v = U_{1,1} \quad (\text{only if } s = \mathsf{L}_N), \tag{4.2}$$

and for each of them $(G_{12})_{r,t}$ has degree at most one. All other entries of G_{12} vanish by (4.1). Hence the coefficient of x^2 in G_{12} vanishes, that is, $BUB = 0$, and Lemma 3.1 gives

$$UB = 0. \tag{4.3}$$

Let $1 \leq t \leq q$ with $t + 1 \leq p$. By (3.1) and (4.2),

$$(UB)_{t,t} = U_{t,t}a_{2t-1} + u_t a_{2t}, \tag{4.4}$$

where $U_{t,t} = 0$ unless $s = \mathsf{L}_N$ and $t = 1$.

Case $s = \mathsf{E}_N$. Here $U_{t,t} = 0$ for all t, so (4.3) and (4.4) give $u_t a_{2t} = 0$, hence $u_t = 0$, for every t in (4.2). Thus $U = 0$.

Case $s = \mathsf{L}_N$, $N = 2$. Here $p = q = 1$, the only possible entry is v, and (4.3) reads $va_1 = 0$. Thus $U = 0$.

Case $s = \mathsf{L}_N$, $N \geq 3$. Now $p \geq 2$, $o_1 = 0$ and $o_2 = 1$. The $(2,1)$ entry of (3.5) is

$$(R - xBU)_{2,1}(x) = x\,\overline{(R - xBU)_{1,2}}(x) = x\,\overline{R_{1,2}} - x^2\,\overline{(BU)_{1,2}}.$$

The left-hand side has degree at most one, so $(BU)_{1,2} = 0$. The first row of B has the single nonzero entry $B_{1,1} = a_1$, and $U_{1,2} = u_1$, so $a_1u_1 = 0$ and $u_1 = 0$. Now (4.3) and (4.4) with $t = 1$ give $va_1 = 0$, so $v = 0$. For $t \geq 2$ we have $U_{t,t} = 0$, and as in the case $s = \mathsf{E}_N$ we get $u_t = 0$. Thus $U = 0$. □

Proof of Proposition 4.1. By Lemma 4.2, $G_{12} = Y + x(RB - BS)$ and (3.6) becomes $G_{12}D_e = 0$. Since the diagonal entries of D_e are nonzero monomials, $G_{12} = 0$, that is,

$$Y = 0, \qquad RB = BS. \tag{4.5}$$

With $U = 0$, identity (3.5) reads $R_{r,t}x^{o_t} = \overline{R_{t,r}}x^{o_r}$ for all r, t. For $r \neq t$ the exponents differ by Lemma 3.2(i), so $R_{r,t} = 0$; for $r = t$ we get $R_{r,r} \in \mathbb{R}$.

Thus $R = \operatorname{diag}(\rho_1, \dots, \rho_p)$ is real, and in the same way (3.7) shows that $S = \operatorname{diag}(\sigma_1, \dots, \sigma_q)$ is real. Comparing the entries of $RB = BS$ at the nonzero entries (3.1) of B gives

$$\rho_t = \sigma_t \quad (1 \le t \le q), \qquad \rho_{t+1} = \sigma_t \quad (1 \le t \le q,\ t+1 \le p).$$

Starting from $\rho_1 = \sigma_1$ and using the two relations alternately we obtain $\rho_1 = \sigma_1 = \rho_2 = \sigma_2 = \cdots$, which reaches ρ_p and σ_q. Hence $F = G = \rho_1 I$ with $\rho_1 \in \mathbb{R}$. Conversely, every real multiple of I lies in $\mathcal{C}_{\mathbb{R}}(W_s)$. □

Proof of Theorem 1.1. Part (a) is Proposition 4.1, and part (b) follows from (a) and Lemma 2.2(i). For part (c), $W_s = wP_s$, where w is a scalar weight on Δ because $s_1 = 0$, and $P_s = TD_sT^*$ is a matrix polynomial with $P_s(x)^* = P_s(x)$ for real x. Proposition 2.3 and part (a) give $\mathcal{D}_0(W_s) = \mathbb{C}I$. □

Proof of Corollary 1.2. Put $w = w_1$. In the first case $w_j = w\,x^{s_j}$ with $s = \mathsf{L}_N$, in the second with $s = \mathsf{E}_N$, and w is s-admissible because each w_j is a scalar weight on the common interval. The weight (1.2) is W_s, and Theorem 1.1 applies. □

Remark 4.3. The proof of Proposition 4.1 compares coefficients and degrees in identities of matrix polynomials, so it does not require that 0 lie in Δ. The scalar weight w enters Theorem 1.1 only through Lemma 3.3 and Proposition 2.3, and neither uses determinacy of the moment problem.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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